

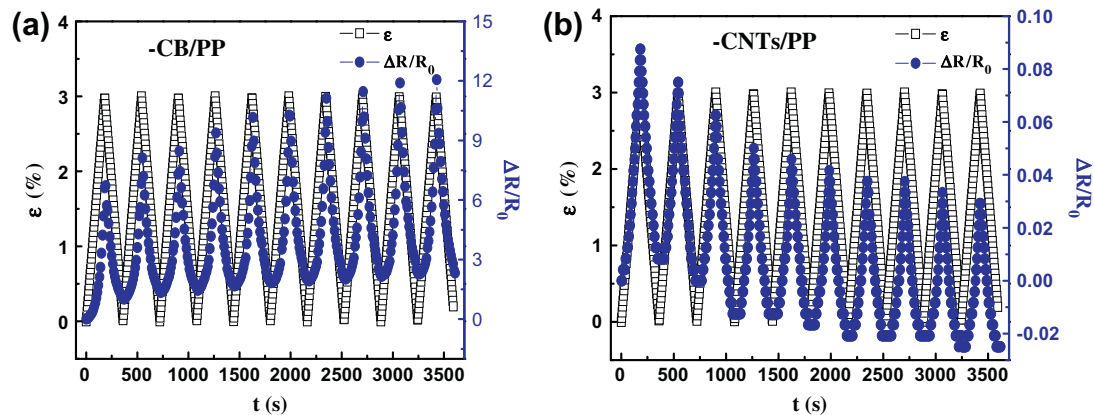


Fig. 8. Relationships of strain and $\Delta R/R_0$ vs. time of (a) CB (7.16 vol.%) /PP and (b) CNTs (3.41 vol.%) /PP composites during 10 extension–retraction cycles. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

The changes of max and min $\Delta R/R_0$ values of CB (7.16 vol.%) /PP and CNTs (3.41 vol.%) /PP during 10 cycles.

Cycle number		1	2	4	6	8	10
CB/PP	max $\Delta R/R_0$	6.72	8.12	9.39	10.28	11.46	12.06
	min $\Delta R/R_0$	1.08	1.34	1.71	1.98	2.24	2.48
CNTs/PP	max $\Delta R/R_0$	0.088	0.075	0.050	0.042	0.038	0.029
	min $\Delta R/R_0$	0.0083	0	−0.013	−0.020	−0.021	−0.025

the applied tensile; and decreases with the retraction of strain. This is because when the sample was dragged, the conductive particles will move with the polymer chains and pulled apart, resulting in an increase of interparticle distance, increasing relative resistance of samples [32]; while retraction, contrary to the stretch, the relative resistance of samples decreased.

More interestingly, for CB/PP, it is found the value of the max $\Delta R/R_0$ increases gradually with increasing the cycle number. The origin is discussed in detail as follows. Take the first extension–retraction cycle as an example (see Fig. 8a), when the strain is increased to the maximum 3%, the breakdown of conductive network plays a dominant role, CB particles motion with PP macromolecules, resulting in large distances between CB particles, i.e. a high $\Delta R/R_0$ value. In the subsequent retraction process, the distances between CB particles decrease gradually due to the releasing of the stress. However, at the end of this cycle, the position change of CB caused by the hysteresis effect resulting from matrix viscoelastic effects is hardly to be eliminated totally. Some vulnerable conductive paths formed by CB particles and aggregates with a low aspect ratio have been destroyed, and a notable increase of the min $\Delta R/R_0$ value to 1.08 in the first cycle is observed. In the extension process of the second cycle, the partially damaged conductive network will be broken further, distances between CB is further increased, leading to an obvious increase of max $\Delta R/R_0$ value (the value increases to 8.12 in the second cycle from 6.72 of the first cycle), so does the min $\Delta R/R_0$ value (it increases to 1.34 from 1.08 in the first cycle). With increasing the cycle number from 1 to 10, CB particles positions are arranged gradually in PP matrix, and increasing max and min $\Delta R/R_0$ values are shown (see Table 2, for simplification, only the first cycle and even number cycles are presented here). The max and min $\Delta R/R_0$ values in the tenth cycle are nearly two times higher than the values in the first cycle, respectively.

In contrast, for the CNTs/PP system, the max $\Delta R/R_0$ value becomes more and more negative during 10 cycles just as reported in literature [33]. The min $\Delta R/R_0$ value also becomes smaller and smaller (see Fig. 8b). It is known conductive network constructed by CNTs with large aspect ratio and entangled structure is very stable. A nice network is beneficial to avoiding the fracture of conduc-

tive path. Moreover, during cycles, the CNTs conductive network is influenced by hysteresis effect resulting from the viscoelastic behavior of CNTs/PP on the molecular level [20], the interaction between CNTs and the viscoelastic nature of the matrix leads to the rearrangement of CNTs and results in a better conductive network along the stretching direction after every cycle. A negative variation of the max and min $\Delta R/R_0$ value is therefore displayed (see Table 2). At the end, the max and min $\Delta R/R_0$ values in the tenth cycle become nearly three times lower than the values in the first cycle.

Based on the results and discussions, a schematic illustration is proposed to explain the mechanism of the different sensing behaviors between CB/PP and CNTs/PP as shown in Fig. 9. The original conductive networks before stretching is shown in Fig. 9a and a'. It can be seen that CNTs with large aspect ratio are arranged in matrix intricately; a nice conductive network by contact of CB particles is also achieved. For the state in tensile process with a high strain as shown in Fig. 9b and b', the breakdown of conductive networks are more predominant than the formation of new ones. For CNTs/PP, the number of conductive paths decreases obviously, while CNTs are rearranged in matrix along the tensile direction [34,35]. During the retraction process, some new conductive paths are formed, especially along the tensile direction. Owing to the hysteresis effect resulting from the viscoelastic behavior of the composites, some of these new conductive paths are preserved, and a much better conductive network is obtained at the end of this cycle (see Fig. 9c), leading to a decreasing max and min $\Delta R/R_0$. For CB/PP in Fig. 9b, due to the relatively weak interaction between the CB particles and aggregates (they have low aspect ratio), after this cycle, though some breakdown of conductive networks will reform again, the number of gaps between CB particles and the gaps' sizes both increase, leading to a increasing max and min $\Delta R/R_0$ as shown in Fig. 8a.

It is evident that the distinction in morphology and microstructure of the two fillers results in the obviously different strain sensing behaviors of the composites. The mechanism of the difference as discussed above of electrical conduction is meaningful for the long-term and cyclic applications of the strain sensors made from CPCs filled by different conductive fillers.

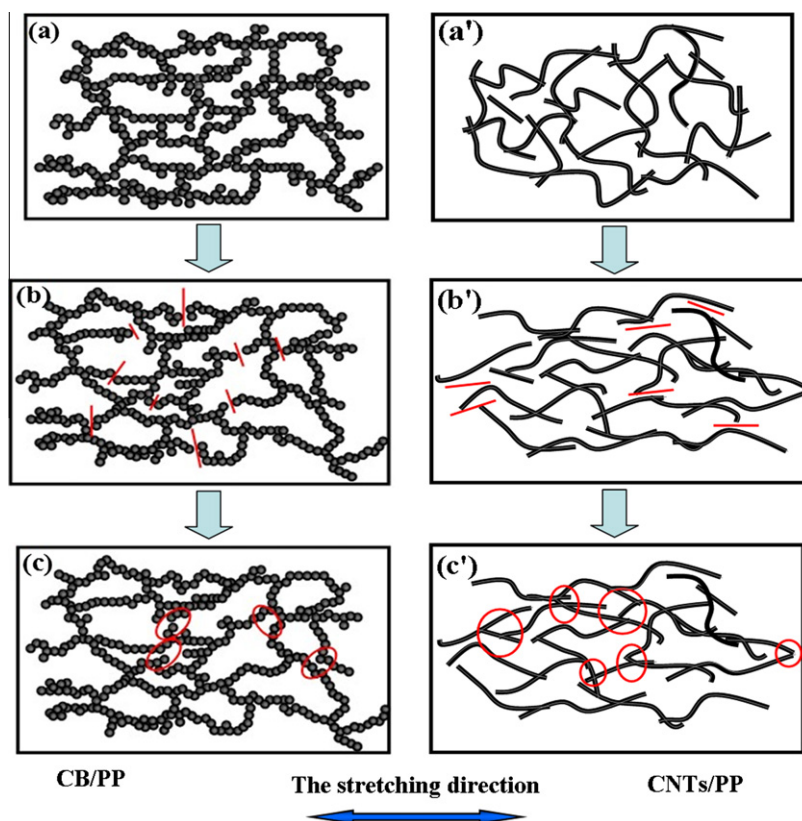


Fig. 9. Schematic illustration of influence of CB and CNTs structure on the strain sensing behavior of CB/PP (a–c) and CNTs/PP (a'–c'): (a and a') initial state; (b and b') the state in the tensile process with a high strain; (c and c') final state of filler distribution. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4. Conclusions

CB/PP and CNTs/PP composites with different filler concentration were prepared through the same processing. SEM observation showed that a nice dispersion of CB and CNTs in PP matrix was achieved. The percolation behaviors of the both CPCs were studied. A percolation threshold *ca.* 3.24 vol.% of CNTs/PP, which is much lower than that of CB/PP, is obtained.

The evolution about the stress and $\Delta R/R_0$ as a function of strain of the two CPCs was investigated. It was found for CB/PP, $\Delta R/R_0$ increases exponentially with the strain. This phenomenon was explained by tunneling mechanism. However, for CNTs/PP, $\Delta R/R_0$ increased in a linear fashion. It is proposed that the main origin of the different trend is the difference in the microscopic morphology of the two fillers, which leads to different conductive networks.

For the sake of evaluating the reproducibility of the strain sensor, the strain and $\Delta R/R_0$ vs. time for both CPCs during 10 cycles were investigated in detail. More interestingly, the values of the max and min $\Delta R/R_0$ in CB/PP and CNTs/PP showed obviously opposite trend with increasing the cycle number, i.e., the positive and the negative effect, indicating a different development of the conductive network. In the end of the tenth cycle, for CB/PP, the max and min $\Delta R/R_0$ values were nearly two times higher than the values in the first cycle; however, for CNTs/PP, the max and min $\Delta R/R_0$ values became nearly three times lower than the corresponding values in the first cycle.

It is suggested upon tension, for CNTs/PP, CNTs were rearranged in matrix along the tensile direction [34,35], the number of conductive paths on the parallel direction of tension increased based on the characteristics of CNTs with a large aspect ratio and the entangled structure. During extension–retraction cycles, a better conductive network was gradually formed due to the hysteresis

effect resulting from viscoelastic behavior of the composites; the decrease of max and min $\Delta R/R_0$ was thus obtained. However, for CB/PP, owing to the relatively weak interaction between the CB particles and aggregates, after several cycles, the number of gaps between CB particles and the gaps' sizes both increased, leading to an increasing max and min $\Delta R/R_0$. A schematic illustration was proposed to explain the mechanism of the different sensing behaviors between CB/PP and CNTs/PP.

The difference in the change of electrical property when two CPCs subjected to cyclical tension mainly depends on the distinction in the microstructure of conductive fillers, the interaction between fillers and matrix and the hysteresis effect resulting from the viscoelastic behavior of composites. These results are useful in designing CB and CNT-based polymer composites with different strain sensing capability to fulfill different requirements.

Acknowledgements

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